Legacy systems to cloud migration: A review from the architectural perspective

Citation: M. H. Hasan *et al.* "Legacy systems to cloud migration: A review from the architectural perspective," *Journal of Systems and Software*, 2023. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0164121223000973

Paper Information: Systematic Literature Review (SLR)

Rating: ⭐ ⭐ ⭐ ⭐

Own Keywords: Background, Cloud Migration, Migration

Reading Progress: Second Pass

Introduction (What is the Paper about?)

- **This paper aims to understand the current state of legacy to cloud migration progress:**
  - The motivation for cloud migration
  - existing cloud migration frameworks and patterns
  - target architecture for the cloud
  - derive post-migration quality issues
- **Focus on to investigate Cloud Migration in general**
  - compare drivers with the resulting concerns
- **Review of the current legacy to cloud migration approaches**
  - encompassing pre-migration, migration and post-migration
- **Implementation point of view**

Methods

- *Systematic Literature Review (SLR)* with 47 selected studies
  - covering pre-migration, migration and post-migration stages
  - maintainability and quality perspective
  - consolidate also previous studies

Scientific Benefit

- **The migration phases are differential well covered**
  - Most studies focus on the planning phase
    - involving a monolith decomposition approach
  - for the migration phase itself only a high-level explanation was given
  - limited studies include the post-migration phase
- **Microservices have been identified as the most targeted architectural pattern for a migration target**
- **Maintainability, scalability and flexibility are the main drivers for cloud migration**
- **Cloud Migration do not fully fulfill the migration drivers**
  - Cloud Migration can satisfy scalability, flexibility, manageability and resiliency but not the main driver of maintainability
  - Security and performance concerns rise after cloud migration
    - performance because of inadequate experience of system architects

- **Migration approaches lack quality consideration**
  - contributing to post-migration quality concerns

Challenges / Future Work

- **Future research might focus on post-migration concerns**

Limitations

Paper Outline and Read Progress

1. [X] Introduction
2. [ ] Related Work
3. [ ] Methods
4. [X] Results: Findings and answer to the RQ's
5. [X] Discussion: Discuss main findings
6. [X] Threats to validity *(No Limitation)*
   - Authors explain why their study is valid
7. [X] Conclusion

Own Comments

- (-) Fist Pass provides not many information about the results of the paper
- (-) Focus on to cloud migration in general without focus on microservices
  - Monolith Migration is a subset of Cloud Migration
- (-) No focus on Service-Interfaces or API-Design